Abstract

We propose a dimensionally consistent phenomenological extension of Jacobson's thermodynamic gravity in which entanglement entropy density S_{ent} ($\mathrm{bit\,m^{-3}}$) enters the Einstein field equations as an additional stress-energy source term. The modified equation, $G_{\mu\nu} = 8\pi G c^{-4}[T_{\mu\nu} + \tilde{\kappa}(c^4/8\pi G k_{\mathrm{B}} \ln 2) S_{\mathrm{ent}} g_{\mu\nu}]$, is dimensionally consistent for any value of the dimensionless coupling $\tilde{\kappa}$. Under the *Planck-scale holographic regulator hypothesis* — that laboratory-scale entanglement entropy couples to geometry through the same Planck-length regulation that governs horizon entropy — we obtain the conjectural estimate $\tilde{\kappa} = -1/4$. This estimate is not derived from first principles; it is a motivated ansatz whose validity is the central open question. We resolve the radial scaling of the predicted acceleration correction to $\Delta a(R) \propto M_{\mathrm{ent}}(< R)/R^2$ for bounded sources, and identify the screening factor α_{screen} and the Planck-scale renormalization of $\tilde{\kappa}$ as the two critical unresolved parameters. The framework is falsifiable: if macroscopic quantum-coherent systems ($\geq 10^6$ entangled qubits) show no anomalous stress-energy at pressure sensitivity $\Delta p < 10^{-6}$ Pa after $\geq 10^3$ independent runs on three or more platforms, then $|\tilde{\kappa}| < 10^{-15}$, ruling out laboratory-scale relevance.

Entanglement Entropy Density as a Gravitational Source: A Dimensionally Consistent Framework and Falsification Criterion

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1 Introduction

Jacobson showed in 1995 that Einstein's field equations follow from the Clausius relation $\delta Q = T\,dS$ applied to local Rindler horizons [1]. Gravity, in this picture, is not a fundamental force but the thermodynamic equation of state of spacetime when entropy scales with horizon area. Verlinde subsequently argued that volume-law entanglement entropy at cosmological scales could produce gravity-like effects without horizons [2].

A natural question follows: does *laboratory-scale* entanglement entropy density contribute measurably to the gravitational source term? The present paper proposes this as a falsifiable phenomenological extension, derives its structural content, and specifies the experimental conditions that would rule it out.

We make no claim that the coupling is derived. We make one claim: the coupling is *testable*.

2 The Framework

2.1 Modified field equation and dimensional consistency

The proposed modification adds an entanglement entropy source term to the Einstein field equations:

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left[T_{\mu\nu} + \tilde{\kappa}\,\frac{c^4}{8\pi G k_{\mathrm{B}} \ln 2}\,S_{\mathrm{ent}}\,g_{\mu\nu}\right], \tag{1}$$

where S_{ent} is the entanglement entropy density in $\mathrm{bit\,m^{-3}}$ (treating "bit" as a dimensionless counting unit) and $\tilde{\kappa}$ is a dimensionless coupling constant. The bit-to-entropy conversion $S_{\mathrm{phys}} = I_{\mathrm{bits}} \cdot k_{\mathrm{B}} \ln 2$ (where $k_{\mathrm{B}} \ln 2 = 9.570 \times 10^{-24}\,\mathrm{J/K}$) ensures dimensional consistency: the entanglement term has the same units as $T_{\mu\nu}$ for any value of $\tilde{\kappa}$.

For a perfect fluid, the active gravitational mass density becomes

$$\rho_{\mathrm{grav}} + \frac{3p_{\mathrm{grav}}}{c^2} = \rho_m + \frac{3p}{c^2} + \frac{3\tilde{\kappa}c^2}{8\pi G k_{\mathrm{B}} \ln 2}\,S_{\mathrm{ent}}. \tag{2}$$

For $\tilde{\kappa} < 0$ and $S_{\mathrm{ent}} > 0$, the entanglement term contributes negative effective pressure: repulsive curvature without exotic matter.

2.2 The conjectural estimate $\tilde{\kappa} = -1/4$

Jacobson's derivation applies the Clausius relation $\delta Q = T_U\,dS$ at a local Rindler horizon, with Unruh temperature $T_U = \hbar a/(2\pi c k_{\mathrm{B}})$ and Bekenstein-Hawking entropy $dS_{BH} = (k_{\mathrm{B}} c^3/4G\hbar)\,dA$.

Under the **Planck-scale holographic regulator hypothesis**, entanglement entropy density $S_{\rm ent}$ contributes an additional entropy

$$dS_{\rm ent} = \frac{S_{\rm ent}}{k_{\rm B}} \cdot \frac{dV}{4\ell_P}, \quad dV = \ell_P\, dA, \tag{3}$$

where $\ell_P = \sqrt{\hbar G/c^3} = 1.616 \times 10^{-35}\,\mathrm{m}$ is the Planck length. Carrying through Jacobson's identification of $\delta Q_{\rm eff}$ with the stress-energy flux yields $T^{\rm ent}_{\mu\nu} = -(c^4/32\pi G) S_{\rm ent} g_{\mu\nu}$, which by comparison with Eq. (1) gives

$$\tilde{\kappa} = -\frac{1}{4}. \tag{4}$$

This is an estimate, not a derivation. Equation (3) imports the Planck-length area-law regulator from horizon physics into a non-horizon volume context. This step has no rigorous justification for systems with volume-law entanglement entropy. The result $\tilde{\kappa} = -1/4$ should be treated as a theoretically motivated ansatz whose correctness is precisely what the experiment tests.

2.3 Radial scaling: resolution of the 1/R vs. 1/R^2 question

In the Newtonian limit, Eq. (1) reduces to a modified Poisson equation for the entanglement potential Ψ:

$$\nabla^2 \Psi = \kappa_{\rm eff}\, \rho_{\rm ent}(R), \quad \kappa_{\rm eff} \equiv \frac{\tilde{\kappa} c^4}{2 k_{\rm B} \ln 2}. \tag{5}$$

Under spherical symmetry, integrating once with regularity at the origin gives

$$\Delta a(R) \equiv -\frac{d\Psi}{dR} = \frac{\kappa_{\rm eff}}{R^2}\, M_{\rm ent}(<R), \tag{6}$$

where $M_{\rm ent}(<R) = 4\pi \int_0^R \rho_{\rm ent}(r)\, r^2\, dr$ is the enclosed entanglement entropy.

The $1/R^2$ factor is universal. A $1/R$ correction requires $M_{\rm ent}(<R) \propto R$, which holds only for sources with $\rho_{\rm ent} \propto 1/r^2$ — a specific extended profile, not a generic prediction. For compact laboratory sources (bounded within radius R_0), the exterior prediction is strictly $\Delta a \propto 1/R^2$ for $R > R_0$. Earlier formulae in this program that assumed a universal $1/R$ scaling encoded a hidden assumption about the source profile and have been corrected.

2.4 The two critical unresolved parameters

The screening factor $\alpha_{\rm screen}$. Environmental decoherence suppresses the coupling: $\tilde{\kappa}_{\rm eff} = -(1/4)\alpha_{\rm screen}$. For macroscopic ^{87}Rb ensembles (cloud radius $\sim$0.1 mm, coherence length $\sim$1 μm), the Gaussian spatial screening factor $\exp[-(R/\xi)^2] \sim e^{-100} \sim 10^{-43}$, which places the predicted signal far below any foreseeable experimental sensitivity. For NISQ-scale systems (50–100 qubits, $R \sim \xi$), $\alpha_{\rm screen}$ may be $\mathcal{O}(10^{-1})$–$\mathcal{O}(10^{-4})$, but this range is heuristic and requires derivation from a Lindblad master equation for the specific system.

Planck-scale renormalization (OP10). The coupling prefactor $c^4/(8\pi k_{\rm B} \ln 2) \approx 5 \times 10^{65}$ in SI units reflects the Planck-scale origin of the derivation. Whether $\tilde{\kappa} = -1/4$ at the Planck scale renormalizes to a far smaller value at laboratory scales is unknown. Both parameters being simultaneously unresolved means the predicted signal magnitude is currently indeterminate. The falsification criterion below is valid regardless.

2.5 The Bianchi identity and matter non-conservation

The contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ applied to Eq. (1) gives

$$\nabla^\mu T^{\rm matter}_{\mu\nu} = -\frac{\tilde{\kappa} c^4}{8\pi G k_{\rm B} \ln 2}\, \nabla_\nu S_{\rm ent}. \tag{7}$$

This is a non-trivial prediction: standard matter is not independently energy-momentum conserved wherever the entanglement entropy density has a spatial gradient. The right-hand

side acts as an effective fifth force on matter proportional to $\nabla S_{\rm ent}$.

Three interpretations are possible, and the theory must ultimately choose one:

(i) **True fifth force**: $\nabla S_{\rm ent}$ drives a composition-independent anomalous acceleration. This is in principle constrained by the Asenbaum et al. 2020 equivalence principle test at $\eta < 10^{-12}$ [4], provided the entanglement entropy density is not identically zero for both test masses.

(ii) **Entanglement-sector exchange**: energy is exchanged between ordinary matter and entanglement degrees of freedom, with total $T^{\rm total}_{\mu\nu}$ (matter plus entanglement field) independently conserved. Under this interpretation, Eq. (7) reflects redistribution, not violation.

(iii) **Incomplete source term**: the modification in Eq. (1) is incomplete, and a term $\delta T^{\rm ent}_{\mu\nu}$ must be added to restore independent matter conservation.

The paper adopts interpretation (ii) as the physically most natural and commits to the following specific statement: the total stress-energy of matter plus the entanglement sector is independently conserved, $\nabla^\mu(T^{\rm matter}_{\mu\nu} + T^{\rm ent}_{\mu\nu}) = 0$, where $T^{\rm ent}_{\mu\nu} = \tilde{\kappa}(c^4/8\pi G k_{\rm B}\ln 2)S_{\rm ent}g_{\mu\nu}$.

Observational constraint on Eq. (7)**.** For two classical test masses (Pt and Ti alloy, as in MICROSCOPE [5]), the entanglement entropy density $S_{\rm ent} \approx 0$ for both, making the right-hand side of Eq. (7) negligible and producing no detectable Eötvös anomaly. MICROSCOPE does not directly constrain $\tilde{\kappa}$. A direct constraint requires a pair of test masses with different $S_{\rm ent}$ – precisely the coherent-versus-decohered comparison in the proposed experiment.

3 Experimental Proposal and Falsification Criterion

3.1 Target regime

The $\alpha_{\rm screen}$ analysis rules out macroscopic neutral-atom ensembles ($N \sim 10^6$ atoms, $R \gg \xi$) as the near-term target. The revised target is a 50–100 qubit NISQ platform (neutral-atom array, trapped ions, or superconducting qubits) where $R \lesssim \xi$ and entanglement can be directly verified.

The experiment compares:

- Branch A: N-qubit state with verified entanglement depth $k \geq 10$, realizing nonzero $M^{\rm tot}_{\rm ent}$.
- Branch B: identical qubit count, same control schedule, entanglement destroyed to produce $M^{\rm tot}_{\rm ent} \approx 0$.

A reproducible differential in any gravitational or inertial observable between branches is attributed to the $S_{\rm ent}$ source term.

3.2 Signal structure

From Eq. (6), the differential acceleration outside a compact source of radius R_0 is

$$\Delta a(R) = -\frac{\tilde{\kappa}_{\rm eff}\, c^4}{8\pi k_{\rm B}\ln 2}\cdot\frac{M^{\rm tot}_{\rm ent}}{R^2}, \quad R > R_0. \quad (8)$$

The signal magnitude is not predicted because $\tilde{\kappa}_{\rm eff}$ requires $\alpha_{\rm screen}$ (unresolved, OP3) and Planck-scale renormalization (unresolved, OP10). The experiment produces a bound on $|\tilde{\kappa}_{\rm eff}|\cdot M^{\rm tot}_{\rm ent}/R^2$, which constrains the product of coupling and entanglement content regardless.

3.3 Falsification criterion

Falsification (F1–F5): The framework is falsified for laboratory-scale relevance if simultaneously: (F1) $\geq 10^6$ genuinely entangled qubits are achieved with depth $\geq 10^3$ verified; (F2) pressure sensitivity $\Delta p < 10^{-6}$ Pa is reached; (F3) $\geq 10^3$ independent null-result runs; (F4) consistent null results across ≥ 3 independent platforms; (F5) no anomalous signal at F2 sensitivity.

Consequence: $|\tilde{\kappa}| < 10^{-15}$.

Current upper bounds from null results in related experiments are listed in Table 1. The atom interferometry bound requires care: the Asenbaum et al. dual-species setup compares different atomic species, while the proposed experiment compares the same system in entangled versus decohered states. The coupling to S_{ent} is entanglement-dependent, not composition-dependent, so the two measurements probe different aspects of any putative coupling.

Table 1: Upper bounds on $|\tilde{\kappa}|$ from existing null results.

Experiment	Bound	Ref.
Gravity-mediated entanglement	$< 3 \times 10^{-9}$	[3][a]
Atom interferometry (equiv. principle)	$\eta < 10^{-12}$	[4][b]
MICROSCOPE (Eötvös parameter)	$\eta < 10^{-15}$	[5][c]

[a] Mapping to $\tilde{\kappa}$ bound to be verified. [b] Probes composition-dependence; requires separate argument for S_{ent}-dependence. [c] Classical test masses ($S_{\text{ent}} \approx 0$); does not directly constrain $\tilde{\kappa}$.

4 Discussion

4.1 Relationship to established programs

This framework is distinct from, but adjacent to, three established programs:

Jacobson 1995 [1]: We extend his thermodynamic derivation from causal horizons to laboratory volumes by adding the holographic regulator hypothesis. The extension is explicit and clearly marked; we do not claim it follows from Jacobson's result.

Verlinde 2017 [2]: Verlinde's emergent gravity addresses galactic scales via volume-law entanglement of dark energy. The present proposal addresses laboratory scales with controlled quantum states. The two are conceptually aligned but make distinct experimental predictions.

ER=EPR [6]: Maldacena and Susskind propose entanglement *is* geometric connectivity. We propose entanglement *sources* stress-energy. These are complementary hypotheses.

4.2 What this paper claims and does not claim

Claims: (i) Eq. (1) is dimensionally consistent for any $\tilde{\kappa}$. (ii) The estimate $\tilde{\kappa} = -1/4$ follows from the holographic regulator hypothesis via thermodynamic algebra. (iii) For bounded sources, the exterior acceleration correction scales as $1/R^2$. (iv) The framework is falsifiable under conditions F1–F5.

Does not claim: (i) The coupling is derived from first principles. (ii) The signal is detectable with current technology. (iii) α_{screen} or the Planck-scale renormalization of $\tilde{\kappa}$ are known. (iv) MICROSCOPE or atom interferometry presently constrain $\tilde{\kappa}$ directly.

5 Open Problems and Path Forward

Three problems must be resolved before the framework can make fully quantitative predictions:

OP3 (Critical): Derive α_{screen} from a Lindblad master equation for a specific NISQ platform. This determines whether the signal is in a detectable regime and what experimental geometry to target.

OP8 (Resolved in framing): We adopt interpretation (ii) (entanglement-sector exchange with total stress-energy conservation) and note that a direct observational test requires a pair of systems with different S_{ent}, not a classical equivalence-principle test.

OP10 (Long-term): Whether $\tilde{\kappa} = -1/4$ at the Planck scale renormalizes to an observable laboratory value is a question in quantum gravity phenomenology. Tools from holographic renormalization group [7] may be applicable.

Progress on any one of these three problems would substantially strengthen the framework. Papers B and C of this research program address the measurement theory prerequisite (Paper B) and the thermodynamic cost structure of entanglement manipulation (Paper C).

6 Conclusion

We have proposed a dimensionally consistent phenomenological framework in which entanglement entropy density sources spacetime curvature. The structural content of the framework — the modified field equation, the radial scaling of the signal, the Bianchi identity consequences, and the experimental falsification criterion — is internally consistent and clearly bounded. The conjectural estimate $\tilde{\kappa} = -1/4$ is stated with explicit epistemic status. The signal magnitude is indeterminate until the screening factor and Planck-scale renormalization are resolved.

The falsification criterion is the framework's primary scientific contribution: it specifies exact, achievable experimental conditions under which laboratory-scale entanglement gravity would be definitively ruled out, making this a falsifiable hypothesis rather than an unfalsifiable metaphysical claim. Either a signal is found, or it is not. Both outcomes advance physics.

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